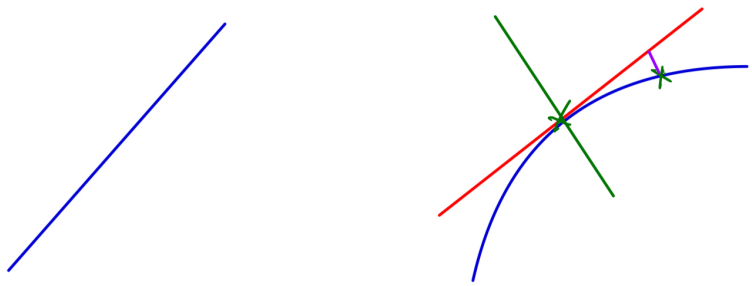
Chapter 2 Properties of Curves

2.1 Curvature

Curvature is a measure of how ”curved” a curve is. A straightline is not curved. A circle of radius r is more ”curved” when r is smaller. Intuitively, the curvature of a curve at a point $\mathbf{x}_0$ is the measure of how much the curve deviates from its tangent line at $\mathbf{x}_0$.



If $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ is a curve that is parametrized by arc-length, then for any $s_0 \in I$, $\boldsymbol{\gamma}'(s_0)$ and $\boldsymbol{\gamma}''(s_0)$ are perpendicular. By Taylor's theorem, in a neighbourhood of s_0,

$$\boldsymbol{\gamma}(s) = \boldsymbol{\gamma}(s_0) + (s - s_0)\boldsymbol{\gamma}'(s_0) + \frac{1}{2}(s - s_0)^2\boldsymbol{\gamma}''(s_0) + (s - s_0)^3\boldsymbol{\Xi}$$

for some vector $\boldsymbol{\Xi}$. Let $\mathbf{N}$ be a unit vector parallel to $\boldsymbol{\gamma}''(s_0)$. Then it is perpendicular to the tangent vector $\boldsymbol{\gamma}'(s_0)$ at s_0. The deviation of $\boldsymbol{\gamma}(s)$ from the tangent line at $\boldsymbol{\gamma}(s_0)$ is captured by $(\boldsymbol{\gamma}(s) - \boldsymbol{\gamma}(s_0)) \cdot \mathbf{N}$. Since $\boldsymbol{\gamma}'(s_0) \cdot \mathbf{N} = 0$, we find that

$$(\boldsymbol{\gamma}(s) - \boldsymbol{\gamma}(s_0)) \cdot \mathbf{N} = \frac{1}{2}(s - s_0)^2\boldsymbol{\gamma}''(s_0) \cdot \mathbf{N} + (s - s_0)^3\boldsymbol{\Xi} \cdot \mathbf{N}.$$

Observe that

$$\boldsymbol{\gamma}''(s_0) \cdot \mathbf{N} = \|\boldsymbol{\gamma}''(s_0)\|.$$

Definition 2.1 Curvature

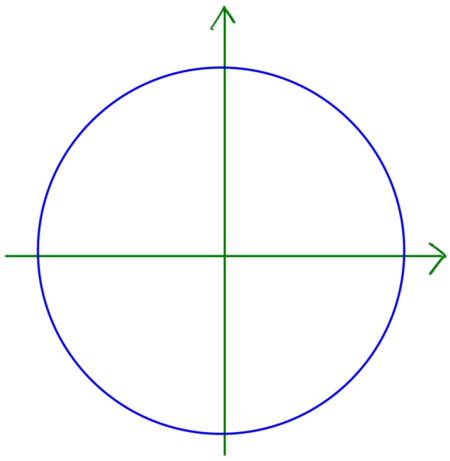
Let $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ be a curve that is parametrized by arc-length. Then the curvature of the curve at $s \in I$ is

$$\kappa(s) = \|\boldsymbol{\gamma}''(s)\|.$$

If the curve is not parametrized by arc-length, we will reparametrize the curve by arc-length to define the curvature.

Example 2.1

Find the curvature of the circle $x^2 + y^2 = r^2$ of radius r.



The following gives the formula to compute the curvature when the curve is not parametrized by arc-length.

Theorem 2.1

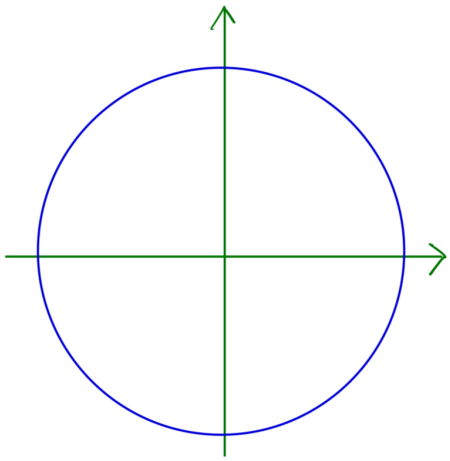
Let $\boldsymbol{\gamma} : I \to \mathbb{R}^3$ be a regular curve in $\mathbb{R}^3$. Then the curvature of the curve at t is

$$\kappa(t) = \frac{\|\ddot{\boldsymbol{\gamma}}(t) \times \dot{\boldsymbol{\gamma}}(t)\|}{\|\dot{\boldsymbol{\gamma}}(t)\|^3}.$$

Example 2.2

For the curvature of the circle $x^2 + y^2 = r^2$ of radius r using the parametrization $\boldsymbol{\gamma} : [0, 2\pi] \to \mathbb{R}^2$,

$$\boldsymbol{\gamma}(t) = (r \cos t, r \sin t).$$

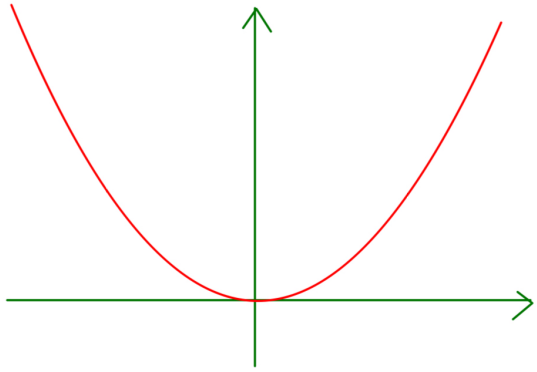


Example 2.3

Let I be an interval and let $f : I \to \mathbb{R}$ be a smooth function. Find the curvature of the curve $y = f(x)$.

Example 2.4

Find the curvature of the curve $y = x^2$.



Theorem 2.2

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ be a regular smooth curve in $\mathbb{R}^n$. Then it has zero curvature if and only if it is a straightline.

2.2 Tangent Vectors and Principal Normal Vectors

Definition 2.2 Unit Tangent Vector and Principal Unit Normal Vector

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ be a curve that is parametrized by arc-length. Then

$$\mathbf{T} = \mathbf{T}(s) = \boldsymbol{\gamma}'(s)$$

is the unit tangent vector at the point s, and

$$\kappa(s) = \|\boldsymbol{\gamma}''(s)\| = \left\| \frac{d\mathbf{T}}{ds} \right\|$$

is the curvature. If $\kappa(s) \neq 0$, then we define the principal unit normal vector $\mathbf{N}(s)$ as

$$\mathbf{N}(s) = \frac{\dfrac{d\mathbf{T}}{ds}}{\left\| \dfrac{d\mathbf{T}}{ds} \right\|} = \frac{\boldsymbol{\gamma}''(s)}{\|\boldsymbol{\gamma}''(s)\|}.$$

The principal unit normal vector $\mathbf{N}(s)$ is a unit vector in the direction of $\dfrac{d\mathbf{T}}{ds}$, and so it is perpendicular to $\mathbf{T}(s)$.

If the curve is not parametrized by arc-length, we will reparametrize the curve by arc-length to define the unit tangent vector and the principal unit normal vector. In the following, we will derive expressions for $\mathbf{T}$ and $\mathbf{N}$ when the curve is not parametrized by arc-length.

By definition,

$$\frac{d\mathbf{T}}{ds} = \kappa(s)\mathbf{N}(s).$$

Proposition 2.3

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ be a parametrized curve. At the point $\boldsymbol{\gamma}(t)$,

$$\mathbf{T}(t) = \frac{\dot{\boldsymbol{\gamma}}(t)}{\|\dot{\boldsymbol{\gamma}}(t)\|}, \qquad \mathbf{N}(t) = \frac{\dfrac{d\mathbf{T}(t)}{dt}}{\left\|\dfrac{d\mathbf{T}(t)}{dt}\right\|}.$$

It is complicated to use the second formula to calculate $\mathbf{N}(t)$.

The following theorem gives us a more practical way to calculate the principal normal vector $\mathbf{N}$ in any parametrization of the curve.

Theorem 2.4

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ be a parametrized curve, and let

$$\mathbf{W}(t) = \ddot{\boldsymbol{\gamma}}(t) - \langle \ddot{\boldsymbol{\gamma}}(t), \mathbf{T}(t) \rangle \mathbf{T}(t).$$

Then $\|\mathbf{W}(t)\| = v(t)^2 \kappa(t)$. If $\kappa(t) \neq 0$,

$$\mathbf{N}(t) = \frac{\mathbf{W}(t)}{\|\mathbf{W}(t)\|}.$$

Corollary 2.5

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^n$ be a parametrized curve with nowhere vanishing curvature, and let $v(t) = \|\dot{\boldsymbol{\gamma}}(t)\|$ be its speed. Then

$$\ddot{\boldsymbol{\gamma}}(t) = a_T(t)\mathbf{T}(t) + a_N(t)\mathbf{N}(t),$$

where

$$a_T(t) = v'(t), \qquad a_N(t) = \kappa(t)v(t)^2$$

are the tangential and normal accelerations.

Example 2.5

Find the unit tangent vector, the principal unit normal vector, and the curvature for the curve $\boldsymbol{\gamma} : \mathbb{R} \to \mathbb{R}^4$,

$$\boldsymbol{\gamma}(t) = (t, t^2, t^3, t^4)$$

at the point $(1, 1, 1, 1)$.

2.3 Plane Curves

In this section, we study curves that lies in a plane. By an isometry of $\mathbb{R}^3$, we can always assume that the curve is in $\mathbb{R}^2$.

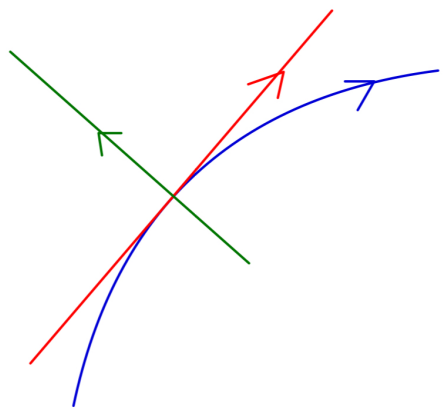
Let $\boldsymbol{\gamma} : I \to \mathbb{R}^2$ be a curve that is parametrized by arc-length. Then

$$\mathbf{T} = \mathbf{T}(s) = \boldsymbol{\gamma}'(s) = (\gamma_1'(s), \gamma_2'(s))$$

is the unit tangent vector at the point s. There are two unit vectors that are perpendicular to $\mathbf{T}$. The signed unit normal vector $\mathbf{N}_s$ is defined to be the one obtained by rotating $\mathbf{T}$ anti-clockwise by $90°$. Thus,

$$\mathbf{N}_s(s) = (-\gamma_2'(s), \gamma_1'(s)) \,.$$

If $\kappa(s) \neq 0$ and $\mathbf{N}(s)$ is the principal normal vector, then



$$\mathbf{N}(s) = \pm \mathbf{N}_s(s).$$

Recall that

$$\boldsymbol{\gamma}''(s) = \kappa(s)\mathbf{N}(s).$$

We define $\kappa_s = \kappa_s(s)$ as

$$\boldsymbol{\gamma}''(s) = \kappa_s(s)\mathbf{N}_s(s).$$

It follows that

$$|\kappa_s(s)| = \|\boldsymbol{\gamma}''(s)\| = \kappa(s).$$

Thus, $\kappa_s(s)$ is known as the signed curvature of the curve at the point s.

Proposition 2.6

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^2$ be a regular plane curve. When $\kappa(t) \neq 0$,

$$\mathbf{N}_s(t) = \operatorname{sgn}(\kappa_s(t))\mathbf{N}(\mathbf{t}).$$

Example 2.6

Show that the circle $x^2 + y^2 = r^2$ with parametrization $\boldsymbol{\gamma} : [0, 2\pi] \to \mathbb{R}^2$,

$$\boldsymbol{\gamma}(t) = (r \cos t, r \sin t),$$

has positive signed curvature.

Example 2.7

Let $f : I \to \mathbb{R}$ be a smooth function. Consider the plane curve $\boldsymbol{\gamma} : I \to \mathbb{R}^2$ with parametrization

$$\boldsymbol{\gamma}(t) = (t, f(t)).$$

Show that its signed curvature at t is positive if and only if $f''(t) > 0$.

Remark 2.1

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^2$ be a regular plane curve whose curvature is nonzero everywhere. Then either κ_s is positive everywhere or negative everywhere. This is an easy consequence of the intermediate value theorem.

Definition 2.3 Turning Angles

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^2$ be a regular plane curve that is parametrized by arc-length. Then there is a smooth function $\varphi : I \to \mathbb{R}$ such that

$$\mathbf{T}(s) = (\cos\varphi(s), \sin\varphi(s)) .$$

It is called the *turning angle* of the curve. The function $\varphi : I \to \mathbb{R}$ is a constant if and only if $\boldsymbol{\gamma} : I \to \mathbb{R}^2$ is a straightline.

Theorem 2.7

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^2$ be a regular plane curve that is parametrized by arc-length, and let $\varphi : I \to \mathbb{R}$ be its turning angle. If $\kappa_s(s)$ is the signed curvature, then

$$\kappa_s(s) = \frac{d\varphi(s)}{ds}.$$

This gives an interpretation of the signed curvature as the rate of change of the turning angle with respect to speed.

Proposition 2.8

Let $\boldsymbol{\gamma} : [a, b] \to \mathbb{R}^2$ be a closed plane curve that is parametrized by arc-length. Then its total signed curvature, defined as

$$\int_a^b \kappa_s(s) ds,$$

is an integer multiple of 2π.

Example 2.8

For the circle $x^2 + y^2 = r^2$ of radius r, the parametrization by arc-length is $\boldsymbol{\gamma} : [0, 2\pi r] \to \mathbb{R}^2$,

$$\boldsymbol{\gamma}(s) = \left(r\cos\frac{s}{r}, r\sin\frac{s}{r}\right).$$

The unit tangent vector is

$$\mathbf{T}(s) = \left(-\sin\frac{s}{r}, \cos\frac{s}{r}\right) = \left(\cos\left(\frac{s}{r} + \frac{\pi}{2}\right), \sin\left(\frac{s}{r} + \frac{\pi}{2}\right)\right).$$

Therefore, the turning angle is

$$\varphi(s) = \frac{s}{r} + \frac{\pi}{2}.$$

The signed curvature is

$$\kappa_s(s) = \frac{d\varphi(s)}{ds} = \frac{1}{r}.$$

Hence, the total signed curvature is

$$\int_0^{2\pi r} \kappa_s(s)ds = 2\pi.$$

Theorem 2.9

Let I be an open interval, and let $k : I \to \mathbb{R}$ be a smooth function. Then there is a plane curve $\boldsymbol{\gamma} : I \to \mathbb{R}^2$ parametrized by arc-length whose signed curvature κ_s is given by k. If $\widetilde{\boldsymbol{\gamma}} : I \to \mathbb{R}^2$ is another such curve, then there is an angle θ_0 and a point (c_1, c_2) such that

$$\widetilde{\boldsymbol{\gamma}}(s) = \begin{bmatrix} \cos\theta_0 & -\sin\theta_0 \\ \sin\theta_0 & \cos\theta_0 \end{bmatrix} \boldsymbol{\gamma}(s) + \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.$$

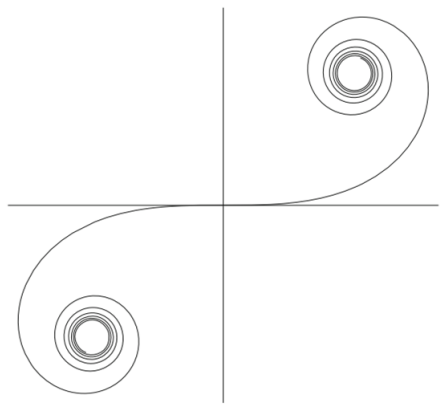
This theorem says that the signed curvature of a plane curve uniquely determines the curve up to a direct isometry.

Corollary 2.10

A plane curve with constant curvature must be part of a straightline or a circle.

Example 2.9

Find a plane curve $\boldsymbol{\gamma} : \mathbb{R} \to \mathbb{R}^2$ parametrized by arc-length whose signed curvature is given by $\kappa_s(s) = s$. Use $s = 0$ as the base point.



2.4 Space Curves

In this section, we study curves in $\mathbb{R}^3$. Recall that when the curvature of a curve is nonzero, we can define the principal unit normal vector $\mathbf{N}(t)$, which is a unit vector perpendicular to the unit tangent vector $\mathbf{T}(t)$.

Definition 2.4 Binormal Vector

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^3$ be a curve that is parametrized by arc-length. If $\kappa(s) \neq 0$, then we can define the principal normal vector $\mathbf{N}(s)$, which is a unit vector perpendicular to $\mathbf{T}(s)$. A vector in $\mathbb{R}^3$ that is perpendicular to both $\mathbf{T}(s)$ and $\mathbf{N}(s)$ is the vector

$$\mathbf{B}(s) = \mathbf{T}(s) \times \mathbf{N}(s).$$

It is known as the binormal vector of the curve at $\boldsymbol{\gamma}(s)$.

Since $\mathbf{T}(s)$ and $\mathbf{N}(s)$ are perpendicular unit vectors, $\mathbf{B}(s)$ is a unit vector that is perpendicular to both $\mathbf{T}(s)$ and $\mathbf{N}(s)$.

The set of vectors $\{\mathbf{T}(s), \mathbf{N}(s), \mathbf{B}(s)\}$ is an orthonormal basis of $\mathbb{R}^3$, and it is a right-handed triplet. It follows that

$$\mathbf{N} \times \mathbf{B} = \mathbf{T}, \qquad \mathbf{B} \times \mathbf{T} = \mathbf{N},$$

and the matrix

$$\mathbf{P} = \left[\mathbf{T}(s) \,\middle|\, \mathbf{N}(s) \,\middle|\, \mathbf{B}(s)\right]$$

is an orthogonal matrix with $\det \mathbf{P} = 1$. The vectors $\mathbf{T}(t), \mathbf{N}(t), \mathbf{B}(t)$ are independent of parametrizations.

Proposition 2.11

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^3$ be a curve in $\mathbb{R}^3$. Then

$$\mathbf{T}(t) = \frac{\dot{\boldsymbol{\gamma}}(t)}{\|\dot{\boldsymbol{\gamma}}(t)\|}, \qquad \mathbf{N}(t) = \frac{\dfrac{d\mathbf{T}}{dt}}{\left\|\dfrac{d\mathbf{T}}{dt}\right\|}, \qquad \mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t).$$

Remark 2.2

We can consider a plane curve $\boldsymbol{\gamma} : I \to \mathbb{R}^2$ as a curve in $\mathbb{R}^3$. If $\kappa(t) \neq 0$, then

$$\mathbf{N}(t) = \operatorname{sgn}(\kappa_s(t))\mathbf{N}_s(t), \qquad \mathbf{B}(t) = \operatorname{sgn}(\kappa_s(t))\mathbf{k}.$$

Example 2.10

For the circle $x^2+y^2 = r^2$ on the plane $z = 0$, it has a parametrization $\boldsymbol{\gamma} : [0, 2\pi] \to \mathbb{R}^3$,

$$\boldsymbol{\gamma}(t) = (r\cos t, r\sin t, 0).$$

Therefore,

$$\mathbf{T}(t) = (-\sin t, \cos t, 0).$$

It follows that

$$\mathbf{N}(t) = (-\cos t, -\sin t, 0), \qquad \mathbf{B}(t) = (0, 0, 1).$$

Example 2.11

Let a be a positive constant, and let b be any constant. Find the triplet $\{\mathbf{T}(t), \mathbf{N}(t), \mathbf{B}(t)\}$ for the curve $\boldsymbol{\gamma} : \mathbb{R} \to \mathbb{R}^3$,

$$\boldsymbol{\gamma}(t) = (a \cos t, a \sin t, bt).$$

Example 2.12

Find the triplet $\{\mathbf{T}(t), \mathbf{N}(t), \mathbf{B}(t)\}$ for the curve $\boldsymbol{\gamma} : \mathbb{R} \to \mathbb{R}^3$,

$$\boldsymbol{\gamma}(t) = (t, t^2, t^3)$$

at the point where $t = 2$.

Definition 2.5 Osculating Plane

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^3$ be a space curve. If the curve has nonzero curvature at the point $\mathbf{x}_0 = \boldsymbol{\gamma}(t_0)$, the osculating plane of the curve at this point is the plane passing through $\mathbf{x}_0$ and spanned by the vectors $\mathbf{T}(t_0)$ and $\mathbf{N}(t_0)$. Hence, it is the plane with equation

$$\mathbf{B}(t_0) \cdot (\mathbf{x} - \mathbf{x}_0) = 0.$$

Example 2.13

Let $\boldsymbol{\gamma} : \mathbb{R} \to \mathbb{R}^3$ be the curve given by

$$\boldsymbol{\gamma}(t) = (3\cos t, 3\sin t, 4t).$$

Find the equation of the osculationg plane at the point $(3, 0, 0)$ on the curve.

Proposition 2.12

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^3$ be a space curve. If the curvature at the point $\boldsymbol{\gamma}(t)$ is nonzero, then $\dfrac{d\mathbf{B}(t)}{dt}$ is parallel to $\mathbf{N}(t)$.

Definition 2.6 Torsion

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^3$ be a space curve parametrized by arc-length. If the curvature at the point $\boldsymbol{\gamma}(s)$ is nonzero, then the torsion of the curve at the point $\boldsymbol{\gamma}(s)$ is defined as the real number $\tau(s)$ such that

$$\frac{d\mathbf{B}}{ds} = -\tau(s)\mathbf{N}(s).$$

Example 2.14

The torsion of a plane curve is 0.

Theorem 2.13 The Frenet-Serret Formulas

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^3$ be a space curve parametrized by arc-length. Assume that the curvature of $\boldsymbol{\gamma}$ is nonzero everywhere. Then

(i) $\dfrac{d\mathbf{T}}{ds} = \kappa \mathbf{N}$

(ii) $\dfrac{d\mathbf{N}}{ds} = -\kappa \mathbf{T} + \tau \mathbf{B}$

(iii) $\dfrac{d\mathbf{B}}{ds} = -\tau \mathbf{N}$

Theorem 2.14

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^3$ be a space curve. If t is a point in I such that the curve has nonzero curvature at the point $\boldsymbol{\gamma}(t)$, then the torsion of the curve at $\boldsymbol{\gamma}(t)$ is given by

$$\tau(t) = \frac{(\dot{\boldsymbol{\gamma}}(t) \times \ddot{\boldsymbol{\gamma}}(t)) \cdot \dddot{\boldsymbol{\gamma}}(t)}{\|\dot{\boldsymbol{\gamma}}(t) \times \ddot{\boldsymbol{\gamma}}(t)\|^2}.$$

Example 2.15

Let a be a positive constant, and let b be any constant. If $\boldsymbol{\gamma} : \mathbb{R} \to \mathbb{R}^3$ is the curve given by

$$\boldsymbol{\gamma}(t) = (a \cos t, a \sin t, bt),$$

find the curvature and torsion of the curve.

Theorem 2.15

Let $\gamma : I \to \mathbb{R}^3$ be a regular space curve with nowhere vanishing curvature. Then the curve lies in a plane if and only if it has zero torsion.

2.5 Fundamental Theorem of Space Curves

Lemma 2.16

Let $\boldsymbol{\gamma} : I \to \mathbb{R}^3$ and $\widetilde{\boldsymbol{\gamma}} : I \to \mathbb{R}^3$ be two space curves that have nonzero curvatures, and assume that $\boldsymbol{\gamma} : I \to \mathbb{R}^3$ is parametrized by arc-length. If there is an orthogonal matrix $\mathbf{P}$ with $\det \mathbf{P} = 1$, and a point $\mathbf{c}$ in $\mathbb{R}^3$ such that

$$\widetilde{\boldsymbol{\gamma}}(s) = \mathbf{P}\boldsymbol{\gamma}(s) + \mathbf{c},$$

then $\widetilde{\boldsymbol{\gamma}} : I \to \mathbb{R}^3$ is also parametrized by arc-length. Moreover, for any $s \in I$,

$$\widetilde{\mathbf{T}}(s) = \mathbf{P}\mathbf{T}(s), \quad \widetilde{\mathbf{N}}(s) = \mathbf{P}\mathbf{N}(s), \quad \widetilde{\mathbf{B}}(s) = \mathbf{P}\mathbf{B}(s),$$

$$\widetilde{\kappa}(s) = \kappa(s) \quad \text{and} \quad \widetilde{\tau}(s) = \tau(s).$$

Exponential of a Matrix

If $\mathbf{A}$ is an $n \times n$ matrix, then $e^{\mathbf{A}}$ is defined as

$$e^{\mathbf{A}} = \mathbf{I}_n + \mathbf{A} + \frac{\mathbf{A}^2}{2!} + \cdots + \frac{\mathbf{A}^k}{k!} + \cdots = \sum_{k=0}^{\infty} \frac{\mathbf{A}^k}{k!}.$$

We have the following.

(a) $\dfrac{d}{dt} e^{t\mathbf{A}} = \mathbf{A} e^{t\mathbf{A}}$.

(b) If $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$ matrices such that

$$\mathbf{AB} = \mathbf{BA},$$

then

$$e^{\mathbf{A}+\mathbf{B}} = e^{\mathbf{A}} e^{\mathbf{B}}.$$

Solutions of System of Homogeneous Linear Differential Equations

Let I be an open interval that contains the point t_0, and let $\mathbf{A} : I \to M_{n\times n}$ be a continuous function defined on I and taking values in the space of $n \times n$ matrices. The system of homogeneous linear differential equation

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}(t)$$

with initial condition

$$\mathbf{x}(t_0) = \mathbf{x}_0$$

has a unique solution.

In the case where $\mathbf{A}(t) = \mathbf{A}$ is a constant, the unique solution is

$$\mathbf{x}(t) = \mathbf{x}_0 e^{\mathbf{A}(t-t_0)}.$$

Theorem 2.17 Fundamental Theorem of the Local Theory of Space Curves

Let I be an open interval, and let $\kappa : I \to \mathbb{R}$ and $\tau : I \to \mathbb{R}$ be smooth functions with $\kappa(s) > 0$ for all $s \in I$. Then there exists a curve $\boldsymbol{\gamma} : I \to \mathbb{R}^3$ parametrized by arc-length so that $\kappa(s)$ and $\tau(s)$ are respectively the curvature and torsion of $\boldsymbol{\gamma} : I \to \mathbb{R}^3$. If $\widetilde{\boldsymbol{\gamma}} : I \to \mathbb{R}^3$ is another such curve, then there is a orthogonal matrix $\mathbf{P}$ with $\det \mathbf{P} = 1$ and a point $\mathbf{c}$ in $\mathbb{R}^3$ such that

$$\widetilde{\boldsymbol{\gamma}}(s) = \mathbf{P}\boldsymbol{\gamma}(s) + \mathbf{c}.$$

Example 2.16

Let α be a positive number and let β be any real number.

(a) Find a space curve $\boldsymbol{\gamma} : \mathbb{R} \to \mathbb{R}^3$ with

$$\kappa(t) = \alpha \quad \text{and} \quad \tau(t) = \beta \qquad \text{for all } t \in \mathbb{R}.$$

(b) Determine all the space curves with $\boldsymbol{\gamma} : \mathbb{R} \to \mathbb{R}^3$ which satisfies

$$\kappa(t) = \alpha \quad \text{and} \quad \tau(t) = \beta \qquad \text{for all } t \in \mathbb{R}.$$

2.6 The Isoperimetric Inequality

If $\boldsymbol{\gamma} : [a, b] \to \mathbb{R}^2$ is a simple closed curve in the plane, it divides $\mathbb{R}^2$ into three parts.

(a) The bounded open set $\mathfrak{D}$ that lies inside the curve, called the interior to the curve.

(b) The unbounded open set $\mathcal{E}$ the lies outside the curve, called the exterior to the curve.

(c) The curve $\mathcal{C} = \boldsymbol{\gamma}([a, b])$.

The closure of $\mathfrak{D}$ is $\overline{\mathfrak{D}} = \mathfrak{D} \cup \mathcal{C}$.

Example 2.17

Let a and b be positive constants, and let $\boldsymbol{\gamma} : [0, 2\pi] \to \mathbb{R}^2$ be the simple closed curve

$$\boldsymbol{\gamma}(t) = (a \cos t, b \sin t).$$

It divides the plane into the three regions:

(a) $\mathfrak{D} = \left\{ (x, y) \;\middle|\; \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} < 1 \right\}$;

(b) $\mathcal{E} = \left\{ (x, y) \;\middle|\; \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} > 1 \right\}$; and

(c) $\mathcal{C} = \left\{ (x, y) \;\middle|\; \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1 \right\}$.

Definition 2.7 Positively Oriented

We say that a simple closed curve $\boldsymbol{\gamma} : [a, b] \to \mathbb{R}^2$ is positively oriented if the signed normal vector $\mathbf{N}_s(t)$ points into the interior $\mathfrak{D}$.

Example 2.18

Let a and b be positive constants, and let $\boldsymbol{\gamma} : [0, 2\pi] \to \mathbb{R}^2$ be the simple closed curve

$$\boldsymbol{\gamma}(t) = (a \cos t, b \sin t).$$

It is positively oriented.

Theorem 2.18 Hopf's Umlaufsatz

The total signed curvature of a simple closed curve in the plane is $\pm 2\pi$.

The proof is beyond our scope.

Line Integrals

If $\boldsymbol{\gamma} : [a, b] \to \mathbb{R}^2$,

$$\boldsymbol{\gamma}(t) = (x(t), y(t))$$

parametrizes a curve $\mathcal{C}$ in $\mathbb{R}^2$, $f : \mathcal{C} \to \mathbb{R}$ and $g : \mathcal{C} \to \mathbb{R}$ are continuous functions, the line integral

$$\int_{\mathcal{C}} (f dx + g dy)$$

is defined as

$$\int_a^b (f(x(t), y(t))x'(t) + g(x(t), y(t))y'(t))\, dt.$$

It is independent of parametrizations.

Theorem 2.19 Green's Theorem

Let $\mathcal{O}$ be an open subset of $\mathbb{R}^2$, and let $\boldsymbol{\gamma} : [a, b] \to \mathbb{R}^2$ be a positively oriented simple closed curve such that its interior $\mathfrak{D}$ has closure $\overline{\mathfrak{D}}$ contained in $\mathcal{O}$. If $f : \mathcal{O} \to \mathbb{R}$ and $g : \mathcal{O} \to \mathbb{R}$ are continuously differentiable functions, then

$$\int_{\mathcal{C}} (f dx + g dy) = \int_{\mathfrak{D}} \left(\frac{\partial g}{\partial x} - \frac{\partial f}{\partial y} \right) dx dy.$$

Theorem 2.20

Let $\boldsymbol{\gamma} : [a, b] \to \mathbb{R}^2$,

$$\boldsymbol{\gamma}(t) = (x(t), y(t))$$

be a parametrization of a positively-oriented simple closed curve $\mathcal{C}$ in $\mathbb{R}^2$, and let $\mathfrak{D}$ be its interior. Then

$$\text{Area of } \mathfrak{D} = \frac{1}{2} \oint_{\mathcal{C}} (xdy - ydx) = \frac{1}{2} \int_a^b \left(x(t)\frac{dy}{dt} - y(t)\frac{dx}{dt} \right) dt.$$

Theorem 2.21 Wirtinger's Inequality

Let $F : [0, \pi] \to \mathbb{R}$ be a continuously differentiable function such that $F(0) = F(\pi) = 0$. Then

$$\int_0^\pi F'(t)^2 dt \geq \int_0^\pi F(t)^2 dt,$$

with equality holds when $F(t) = C \sin t$ for some constant C.

Theorem 2.22 Isoperimetric Inequality

Let $\boldsymbol{\gamma} : [a, b] \to \mathbb{R}^2$ be a parametrization of a simple closed $\mathcal{C}$, and let $\mathfrak{D}$ be its interior. Denote by $\ell(\mathcal{C})$ the arc-length of $\mathcal{C}$, and by $\mathcal{A}(\mathfrak{D})$ the area of $\mathfrak{D}$. Then we have the isoperimetric inequality

$$\mathcal{A}(\mathfrak{D}) \leq \frac{\ell(\mathcal{C})^2}{4\pi}.$$

The equality holds if and only if $\mathcal{C}$ is a circle.

The inequality says that among all the simple closed curve with the same arc-length, the circle encloses the largest area.